

KORT Patient Management System

Document Type: Complete Operations, Access, and Workflow Manual

Application: KORT Assest Managment System

Module: Patient Care / Patient Management

Version: 1.0

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This document is aligned to the implemented Laravel/Inertia module and its current permissions, routes, workflows, and audit behavior.

- Patient Registry
- Admission Workflow
- Diagnosis + Prescriptions
- Print/PDF
- Audit Compliance

1. Module Objective

The Patient Management System extends KORT into a hospital-ready clinical workflow where staff can register patients, admit them, assign doctor/ward/bed, record visits and diagnosis, issue prescriptions, and maintain a permanent clinical history.

Core objective in one line:

Search or register patient -> Admit if needed -> Doctor documents visit and diagnosis -> Create prescription -> Print/Download PDF -> Preserve full history and audit logs.

2. Scope and Integration

- Built inside the existing KORT Laravel application; no change to Asset, Inventory, Procurement, and core UI architecture.
- Uses existing authentication and role-permission framework (Spatie Permission).
- Uses existing activity/audit logging conventions.
- Uses existing blue/white modern Inertia UI flow.
- Supports OPD, Emergency, and admitted follow-up workflows.

3. Data Design (Clinical Record Structure)

The module stores patient care through normalized entities:

Table	Purpose	Key Identifiers
patients	Master demographic profile	patient_number, cnic
patient_admissions	Inpatient admissions and status lifecycle	admission_number, status
patient_visits	Clinical visit episodes	visit_number, visit_type
patient_diagnoses	Doctor diagnosis and severity	visit_id, follow_up_date
patient_prescriptions	Prescription header and instructions	prescription_number
patient_prescription_items	Medicine line items	prescription_id

Table	Purpose	Key Identifiers
patient_documents	Uploaded reports/documents	patient_id, visit_id

4. Role and Rights Matrix

Patient module permissions are enforced per role. Below is the practical access summary:

Role	Patient Module Rights	Typical Responsibility
Super Admin	Full patient permissions including delete, discharge, print and admin actions	Governance, overrides, compliance
Hospital Admin	Full operational access (patient/admission/visit/diagnosis/prescription)	Operational supervision, user control
Doctor / Consultant	View/search/create/edit patient, create/edit diagnosis, create/print prescription, discharge	Clinical decisions and treatment
Receptionist	Create/search/edit patient, create admissions, view admissions	Registration and front-desk admission
Nurse Supervisor	View patients/admissions/visits; edit visit notes/vitals (if needed by workflow)	Bedside support and monitoring
Staff Nurse / Clinical User	View patients/admissions/visits; edit visit notes/vitals	Clinical support tasks
Burn Center Manager / Department Head	Broad view + create patient, admission, visit, diagnosis, prescription and print	Department-level control

4.1 Patient Permission Keys

Permission	Meaning
patient.view	View patient records and profile/history pages
patient.create	Register new patients
patient.edit	Edit demographics/contact/medical details
patient.delete	Archive patient record
patient.search	Search by CNIC, number, name, phone
patient-admission.view / create / edit / discharge	Admission lifecycle control
patient-visit.view / create / edit	Visit entries and updates
patient-diagnosis.view / create / edit	Clinical diagnosis control
patient-prescription.view / create / print	Treatment writing and print/PDF

5. User Management and Password Authority

5.1 Who Can Add New Users?

New staff users can be added from **Administration -> Users** by roles having **users.create**. In the current matrix, this is primarily **Super Admin** and **Hospital Admin**.

5.2 Who Can Change User Access and Roles?

User profile updates require **users.update**. Role assignment changes are controlled by **users.assign-roles** and role permissions by **roles.update**.

5.3 Who Can Change Passwords?

Scenario	Allowed Actor	Control Rule
User changes own password	Any authenticated user	Profile password update requires current password confirmation
Admin sets/resets another user password	Roles with users.update via User edit form	Password field can be updated on user edit; blank leaves unchanged
Forgot-password flow	User owning email account	Standard reset link workflow (depends on mail setup)

6. Operating Procedures

6.1 Register New Patient (Reception / Admin)

- Step 1:** Open **Patient Care -> Patient Search** and search first using CNIC, patient number, name, or phone.
- Step 2:** If no existing record matches, go to **Patients -> New Patient**.
- Step 3:** Enter demographics, emergency contact, allergies, medical history, optional photo.
- Step 4:** Save; patient number is auto-generated when not manually entered.

CNIC is validated for Pakistani format (#####-#####-# or 13 digits) and stored in normalized format to prevent duplicates.

6.2 Fast Search Logic

- Exact CNIC or exact patient number: opens patient profile directly.
- Name/phone query with multiple matches: shows result list for selection.
- Search is available on Patient List and dedicated Patient Search page.

6.3 Patient Admission Workflow

- Step 1:** Open patient profile.
- Step 2:** Click Admit Patient.
- Step 3:** Enter admission date/time, department, ward/room/bed, attending doctor, reason, initial condition.
- Step 4:** Save admission with status **admitted**.

Admission status lifecycle supports: **admitted, discharged, transferred, cancelled**.

6.4 Doctor Queue and Visit Handling

- Doctors open **Patient Care -> Doctor Queue** to see admitted patients assigned to them.

- From patient/admission context, doctor creates diagnosis entry which automatically creates linked visit record.
- Visit captures complaint, vitals, notes, visit type, date/time, doctor, and optional admission link.

6.5 Diagnosis Workflow

Step 1: Open Create Diagnosis for patient.

Step 2: Fill visit details and chief complaint.

Step 3: Enter diagnosis, clinical notes, severity, and follow-up date.

Step 4: Save to permanently append diagnosis and visit history.

6.6 Prescription Workflow

Step 1: After diagnosis save, open Create Prescription.

Step 2: Add one or more medicine rows (name, dosage, frequency, duration, instructions).

Step 3: Add general instructions and printable notes.

Step 4: Save; prescription is linked to patient and visit with unique prescription number.

7. Prescription Print and PDF

Authorized users with **patient-prescription.print** can:

- Open browser-print view ([Print Prescription](#))
- Download official PDF ([Download PDF](#))

Prescription output includes:

- Hospital/system name
- Patient name, patient number, CNIC, age/gender
- Date/time and doctor name
- Diagnosis summary and follow-up date
- Medicine table and instructions
- Doctor signature area

8. Patient Profile and History Design

The profile page is a complete clinical dashboard per patient:

- Demographics and contact information
- Allergies and medical history
- Current admission status
- Past admissions list
- Visit history
- Diagnosis history
- Prescription history with print/PDF access
- Uploaded documents and notes

The History page renders a timeline-style longitudinal patient journey for clinical and compliance review.

9. Audit and Compliance Controls

All critical patient events are activity-logged, including:

- Patient created / updated / archived
- Admission created / updated / discharged
- Visit created / updated
- Diagnosis created / updated
- Prescription created / printed / PDF downloaded
- Patient document uploaded

This logging supports traceability, accountability, and post-incident review.

10. End-to-End Working Design (Operational Blueprint)

- Front Desk Layer:** Search-before-create, patient registration, admission initiation.
- Clinical Layer:** Doctor queue, visit capture, diagnosis, medication planning.
- Treatment Output Layer:** Prescription print and PDF generation for bedside and pharmacy use.
- History Layer:** Longitudinal timeline and profile retrieval for follow-up continuity.

11. Quick Route Map (Patient Module)

Area	Representative Route Name(s)
Patient list/profile	patients.index, patients.show, patients.history
Search/queue	patients.search, patients.queue
Admissions	patients.admissions.create/store/show/update/discharge
Diagnosis & visits	patients.diagnoses.create/store/edit/update, patients.visits.update
Prescriptions	patients.prescriptions.create/store/show/print/pdf
Documents	patients.documents.store

12. Testing and Operational Verification

- Feature tests cover patient search redirect behavior (CNIC and patient number exact match).
- Feature tests validate diagnosis + visit creation and prescription + print/PDF flow.
- Seeder provides a sample patient, admission, diagnosis, prescription, and document linkage for demonstrations.

13. Implementation Notes for Admin Team

- Use role-based access only; avoid assigning broad rights to non-admin users.
- Ensure CNIC entry standardization at front desk.
- Use discharge summary consistently for inpatients.
- Review audit logs routinely via Security section.
- Keep doctor and ward assignment accurate to maintain queue quality.

14. Conclusion

The Patient Management System is now integrated as a complete hospital workflow layer in KORT with controlled access, durable medical records, printable treatment output, and auditable operations suitable for professional healthcare use.

End of document.